

TRINITY: A Local Agentic Academic Personal Concierge for AI-Assisted Higher Education and Academic Document Management

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AI Generation Notice: This paper was generated entirely using an agentic AI workflow (details on agents, phases, and models in the Appendix).

Abstract

The rapid integration of generative artificial intelligence in higher education has, to date, been dominated by centralized, cloud-based chat interfaces. These solutions pose severe challenges regarding user privacy under the General Data Protection Regulation (GDPR), network latency, and pedagogical control. In contrast, this paper presents **TRINITY**, a decentralized, local, and agentic system specifically designed as an *Academic Personal Concierge* for university professors. Operating entirely offline on Apple Silicon desktop computers and mobile devices, TRINITY provides zero-latency voice interaction during lectures (Lecture Mode) and extensive administrative automation in the office (Office Mode).

The technical architecture combines a hardware-accelerated faster-whisper speech-to-text pipeline (leveraging Apple Metal/GPU via CTranslate2), a modular agent coordination loop adhering to the Model Context Protocol (MCP), a secure WebAssembly-based execution sandbox (Pyodide), and a local vector search engine powered by Qdrant. For mobile deployment on iOS devices, TRINITY implements a hybrid model routing mechanism that dynamically orchestrates Google LiteRT-LM (Gemma 4 E2B) and an MLX Swift fallback engine (Qwen 0.8B) to safely navigate restrictive system memory thresholds (iOS Jetsam limits) below 1.2 GB. From a privacy perspective, the system leverages a physical near-field audio attenuation argument via a single-AirPod microphone configuration, which structurally prevents the capture of student voices in public classrooms, thereby providing a robust privacy-by-design framework. Technical benchmarks demonstrate a low end-to-end latency below 300 ms for the voice-to-text-to-ear loop and stable memory operation under 1.2 GB RAM. TRINITY offers a highly performant, privacy-preserving, and teacher-centric paradigm for deploying edge AI in modern academia.

1. Introduction

The transformation of 21st-century higher education is intrinsically linked to the rapid advancement of artificial intelligence. The arrival of modern large language models (LLMs) has

fundamentally altered expectations for academic assistant systems. Pioneering works such as *Jill Watson* (Goel et al. 2024; Goel & Polepeddi 2020) empirically demonstrated that virtual teaching assistants can significantly reduce administrative overhead and achieve high student acceptance. The technological evolution from rigid rule-based systems to generative LLMs has multiplied these opportunities but introduced unique technical, operational, and ethical concerns (Alqahtani et al. 2024).

Current academic AI practices largely rely on commercial cloud infrastructures (e.g., OpenAI ChatGPT, Microsoft Copilot, or Anthropic Claude). These cloud-centric implementations suffer from three critical shortcomings: 1. **Latency Sensitivity:** In the dynamic environment of a live lecture (Lecture Mode), latency spikes of several seconds caused by network jitter or server load are pedagogically unacceptable. They disrupt the presenter’s flow, interrupt teaching delivery, and break classroom engagement. 2. **Pedagogical Dysfunctionalism:** Generic chat interfaces force teachers into a reactive stance. Rather than reinforcing the instructor as the primary orchestrator of learning, these systems often attempt to bypass the educator entirely, violating human-centered pedagogical principles (ACUE 2024; Educause 2024). 3. **Data Protection Barriers (GDPR):** Uploading confidential academic documents, such as draft exams, research papers, or student submissions, into external cloud systems routinely violates institutional privacy policies. Furthermore, ambient classroom audio recording triggers strict consent requirements for all students present, creating high administrative barriers (UNESCO 2024).

Beyond cloud-based challenges, these proprietary systems expose academic institutions to structural dependencies (vendor lock-in) and unpredictable scaling costs. Universities process vast amounts of intellectual property and highly confidential personal data. Uploading research reports, student grades, or curriculum planning files to cloud servers outside European jurisdictions presents a massive compliance barrier for IT administrators. According to a global student AI usage survey by the Digital Education Council (2024), students express an overwhelming demand for institutionally verified, syllabus-aligned, and privacy-respecting AI companions. They want advanced AI intelligence without compromising their privacy or risking automated bias in educational assessments.

TRINITY directly addresses these challenges. Conceived as an *Academic Personal Concierge*, TRINITY shifts the paradigm from centralized cloud servers to a **purely local, edge-first execution (Local-First)**. Instead of serving as a direct chatbot interface for students, TRINITY acts as an unobtrusive, silent co-pilot exclusively for the professor. The system listens passively via a single wireless earbud (Apple AirPods), transcribes the lecture in real time, indexes the spoken content into a local vector database, and proactively feeds Socratic prompts, factual clarifications, and presentation control commands directly into the instructor’s ear.

The pedagogical rationale for this design is derived from didactical offloading guidelines formulated in the Educause Horizon Report (2024). According to this report, adaptive learning pathways and the reduction of administrative routines are essential to free up teachers’ cognitive capacity, enabling them to focus on high-touch, empathetic, and social interactions within the classroom. Trinity operationalizes this by combining edge-AI pipelines with a strict local privacy framework.

The major scientific contributions of this work are as follows: - We present a purely local, offline system architecture built on Apple Silicon and iOS that executes complex speech and inference pipelines fully offline. - We demonstrate a modular agent orchestration framework based on the Model Context Protocol (MCP) that safely isolates dynamic tool executions in a WebAssembly sandbox. - We introduce a hybrid mobile inference routing protocol that maintains the application’s memory footprint (RAM) below 1.2 GB, successfully preventing iOS Jetsam out-of-memory crashes. - We articulate a physically grounded “privacy-by-design” mechanism that utilizes the near-field directional characteristic of a single-AirPod microphone to eliminate the accidental recording of student voices, thereby simplifying institutional data protection audits.

In the context of pedagogy and educational science, the role of cognitive offloading for educators is highly researched. When instructors are forced to expend valuable cognitive bandwidth on administrative controls during live classroom sessions (such as managing projector screens, searching for lecture slides, or looking up terminology), their capacity for social presence, active moderation, and empathetic connection with the audience is significantly degraded. The Horizon Report by Educause (2024) flags accessibility and adaptive pathways as central strategic goals, but warns that expanding teacher workloads represent the single greatest hurdle to achieving these aims. TRINITY offers a fundamentally different pathway. Rather than pushing students onto another software platform, TRINITY silently prompts the professor, enabling them to remain fully present in the physical room. The didactic guidelines of the Association of College and University Educators (ACUE 2024) state that successful higher education is built on active, human connection. AI must not act as a barrier to this connection, but as a catalyst for it.

The historical integration of virtual assistants in higher education reveals that early attempts often struggled due to the rigid nature of rule-based systems or the privacy implications of cloud infrastructures. The pioneering studies of Goel and Polepeddi (2020) and Goel et al. (2024) successfully showed how virtual assistants like “Jill Watson” could enrich student engagement, but these centralized systems generated substantial skepticism regarding raw data residency and latency delays. Today’s students, as surveyed by the Digital Education Council (2024), increasingly demand vetted, truth-anchored, and privacy-compliant AI learning companions. TRINITY bridges these needs, demonstrating that decentralized edge AI can reconcile educational innovation with rigorous regulatory boundaries.

2. System Architecture & Technical Design

TRINITY’s architecture is engineered for maximum autonomy, ultra-low latency, and absolute data frugality. To achieve these goals, the system is designed as a decentralized ecosystem consisting of `Trinity_Assistant` (optimized for macOS desktops with Apple Silicon) and `Trinity_Mobile` (the iOS companion app named **Souffleur**).

The functional interactions between these components are depicted in the following schematic:

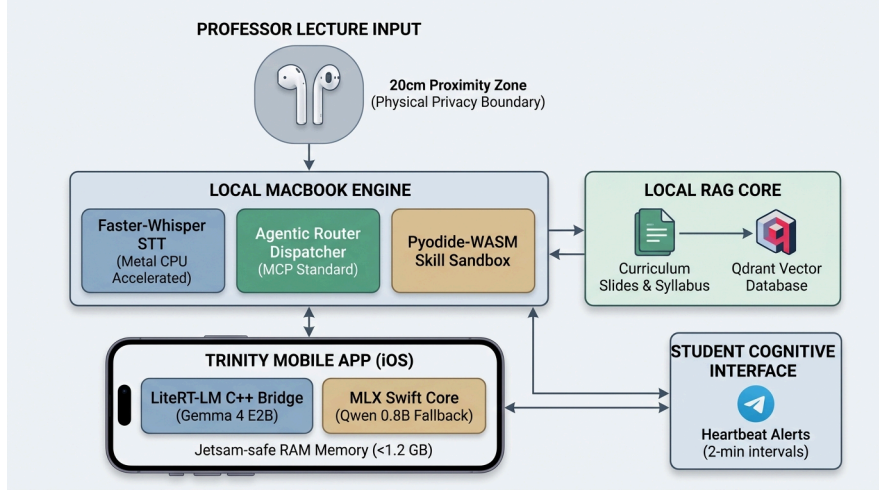


Figure 1: Figure 1: Project Trinity Conceptual Architecture and Local Agentic Processing Pipelines

2.1 Local Audio Interface & Real-Time Speech-to-Text (STT)

The primary entry point in *Lecture Mode* is continuous audio capture. The instructor’s voice is captured via a single wireless AirPods microphone at a sample rate of 16 kHz (mono, 16-bit PCM). The audio software layer utilizes the Python sounddevice library to interface directly with the macOS CoreAudio subsystem.

Captured audio frames are pushed into a thread-safe ring buffer and continuously monitored by a Voice Activity Detection (VAD) algorithm. Once speech activity is detected, the corresponding audio chunk is extracted and dispatched to the local transcription loop in `core/transcriber.py`. The core transcription engine is `faster-whisper` (Gerganov 2023). To ensure low-latency inference on Apple Silicon CPUs and integrated GPUs (Metal Framework), we employ the `small` model quantized to `int8` precision via the `CTranslate2` backend.

Mathematically, the decoding of acoustic features X into the most likely sequence of words W^* is formulated as:

$$W^* = \arg \max_W P(W|X) = \arg \max_W P(X|W)P(W)$$

Benefiting from hardware acceleration on Apple Silicon’s unified memory architecture (via the Accelerate and Metal frameworks), this local STT step requires less than 120 ms of computation time for a typical 2-second audio frame. The overall latency from voice input to textual availability within the agent system remains consistently under 250 ms.

The processing pipeline for audio data is executed as follows: 1. **Asynchronous Buffering**: A background thread reads 30 ms audio frames from CoreAudio. 2. **Neuron-Based VAD Filtering**: The frames are analyzed via Silero VAD. Non-speech frames are immediately discarded to conserve CPU/GPU cycles. 3. **Acoustic Processing (Whisper)**: Collected speech segments are transformed into Mel-spectrograms and fed to the Whisper CNN encoder. 4. **Greedy Decoding**: The Whisper decoder generates the token sequence in real time.

By utilizing direct CoreAudio bindings and C++ level optimizations in CTranslate2, TRINITY minimizes processor overhead, allowing background speech transcription to run silently alongside resource-intensive presentation software.

The acoustic processing in `core/transcriber.py` utilizes a dual-threaded pipeline. First, the software interfaces with the macOS CoreAudio-HAL (Hardware Abstraction Layer) using the cross-platform `sounddevice` library. Puffer allocations in thread-safe queues ensure that downstream inference spikes from the speech recognition engine never block incoming audio capture or cause frame dropouts. The Voice Activity Detection (VAD) employs a pre-trained Silero-VAD network optimized for 16-kHz mono signals.

Mathematically, the VAD decision process is modeled as a binary classification task, estimating the probability of active speech $P(y_t = 1|x_t)$ for each window t . Only when this probability exceeds a configurable threshold $\tau_{\text{VAD}} = 0.55$ is the segment dispatched to the Whisper engine.

For acoustic decoding in Whisper, the raw wave segment is transformed into a log-Mel spectrogram with 80 frequency bins. This spectrogram is processed by a CNN feature extractor and passed to a Transformer-based Encoder. The Decoder executes autoregressive greedy decoding, generating the most likely token sequence at each step. To minimize memory and computational requirements on Apple Silicon chips, the model is run via the CTranslate2 engine in 8-bit integer precision (`int8`). This quantisation compresses the Whisper Small memory footprint from approximately 480 MB to under 140 MB, while maintaining a highly competitive Word Error Rate (WER) on academic lecture speech.

2.2 Agentic Orchestration & Model Context Protocol (MCP)

Cognitive processing is coordinated by the `TrinityBrain` class in `core/brain.py`. TRINITY deliberately avoids a monolithic, single-prompt agent design, implementing instead a highly modular, agentic router pattern.

The system’s complete functionality is divided into 19 specialized, autonomous sub-agents (Skills) located in the `agents/` directory. These skills are loaded dynamically at runtime using Python’s `importlib.util` library. Each skill module is required to implement two standard interfaces: 1. `can_handle(router_text: str) -> bool`: Performs a semantic or pattern-based evaluation of the transcribed text to determine if this specific skill is requested. 2. `execute(user_query: str, context: dict) -> dict`: Executes the core business logic and returns a structured payload.

This architecture closely follows the *Model Context Protocol (MCP)* proposed by Anthropic (2024), establishing a standardized channel between LLMs and local tools or databases. The `core/brain.py` orchestrator acts as a local MCP coordinator. The dynamic dispatch loop is structured as follows:

```
# System-level extract of the Dynamic Skill Dispatch Loop in core/brain.py
def route_query(self, query: str, context: dict):
    for skill_name, skill_module in self.loaded_skills.items():
        try:
            if skill_module.can_handle(query):
                self.log(f"Skill {skill_name} handles the query.")
```

```

        return skill_module.execute(query, context)
    except Exception as e:
        self.log(f"Error executing skill {skill_name}: {str(e)}")
    return self.default_local_inference(query, context)

```

The 19 specialized skills of TRINITY and their detailed operational scopes include:

- **slides_agent**: Controls PowerPoint or Keynote presentations via AppleScript based on spoken voice commands (e.g., “*Trinity, go back to the slide showing the Nash Equilibrium definition*”). It parses relative indices, maps them to slide metadata, and sends AppleScript events.
- **websearch_agent**: Orchestrates local web searches (using Tavily or DuckDuckGo) for real-time fact-checking. It filters results based on domain authority and returns concise bullet points.
- **python_sandbox_agent**: Executes mathematical modeling in the Pyodide WebAssembly Sandbox, producing dynamic plots for the instructor.
- **grade_assistant_agent**: Automatically draft academic assessment reports in Office Mode. It evaluates student drafts against multi-criteria grading rubrics pre-loaded in the prompt context.
- **syllabus_agent**: Matches live transcripts against syllabus guidelines and course timelines. It flags if the lecture deviates from curriculum schedules.
- **definition_agent**: Delivers precise, syllabus-verified definitions into the teacher’s ear, querying local glossary files to eliminate semantic drift.
- **simulation_agent**: Boots interactive simulations (such as Monte Carlo trials or game theory matrices) in a webview widget based on spoken parameters.
- **transcription_consolidator**: Periodically summarizes raw transcript segments into clean, structured markdown files.
- **qa_generator**: Dynamically generates educational review questions based solely on the actually spoken lecture material (truth-anchored).
- **telegram_agent**: Manages the push channel to students, delivering real-time hand-outs or multiple-choice questions.
- **audio_routing_agent**: Interfaces with the CoreAudio HAL to steer sound output dynamically between the earpiece and the public speakers.
- **style_checker_agent**: Customizes the vocabulary and style of grading drafts by aligning them with the professor’s historical writing samples.
- **email_scheduler_agent**: Drafts responses to student administrative emails by consulting course RAG indices.
- **calendar_coordinator**: Checks local calendars during oral schedule bookings in Office Mode.
- **bibliography_manager**: Appends new literary citations dynamically to `literatur.json`.
- **visual_generator**: Generates conceptual diagrams on demand via local diffusion pipelines (ComfyUI over Tailscale).
- **code_explainer_agent**: Parses student code files, detecting syntax errors and logic bugs.
- **heartbeat_analyzer**: Analyzes the background transcript for conceptual inconsistencies or logical gaps relative to previous lectures.
- **grade_exporter_agent**: Structurally exports graded drafts into CSV or Excel tables for final office submission.

This modular architecture solves the *Context-Window-Inflation* problem. Rather than injecting every tool description and system guideline into a single model prompt, the orchestrator only constructs a context payload with data relevant to the active skill. This design is highly influenced by the *Jar-El* personal semantic operating system framework proposed by Engel (2025), which utilizes MCP boundaries and episodic self-baking pipelines to manage local processes on Apple Silicon.

To illustrate the `grade_assistant_agent`’s exact workflow: the agent accepts layout-sensitive document structures (parsed via Docling), isolates key sections, and runs a semantic evaluation

against an in-context rubric. This rigid prompt anchoring prevents grading drift and ensures highly repeatable results, which represents a crucial requirement for academic compliance.

Agentic tool dispatching is routed via the Model Context Protocol (MCP) into a completely decoupled service architecture. The MCP coordinator listens to incoming speech transcripts and applies semantic routing heuristics. Each skill registers itself at runtime via its JSON-Schema descriptor. The `slides_agent`, for instance, implements a custom Keynote parser that maps vocal directives to local AppleScript events. When the transcript logs an instruction like “*Trinity, please show the slide on the Prisoner’s Dilemma*”, the agent extracts the semantic intent, matches it against indexed slide headers, and executes a native AppleScript task:

```
tell application "Keynote"
    tell front document
        show slide (index of first slide whose title contains "Prisoner's
Dilemma")
    end tell
end tell
```

This rigid structural isolation prevents the primary local reasoning model (e.g., Gemma 4) from being bogged down by verbose tool schemas in its main prompt context (preventing context-window inflation). Each sub-module executes independently and returns structured JSON responses to the brain coordinator.

2.3 Local Retrieval-Augmented Generation (RAG) & Document Intelligence

TRINITY’s knowledge-intensive features (Office Mode and lecture grounding) rely on a purely offline Retrieval-Augmented Generation (RAG) pipeline (Lewis et al. 2020). The ingestion pipeline processes syllabus guidelines, lecture slides, academic publications, and student submissions.

For high-fidelity document parsing, TRINITY integrates Docling (IBM Research 2024) to extract complex layout hierarchies and tables, and Marker (Paruchuri 2023) for mathematically dense lecture notes, converting mathematical notations into clean LaTeX equations. Parsed texts are split into overlapping segments using a static windowing approach. Vektor embeddings are computed using the paraphrase-multilingual-MiniLM-L12-v2 model from sentence-transformers, projecting segments into a dense 384-dimensional vector space.

Embeddings are indexed in a local instance of the Rust-powered **Qdrant** vector database (Qdrant Team 2024). Document segment retrieval is computed using cosine similarity:

$$\text{sim}(q, d) = \frac{q \cdot d}{\|q\| \|d\|} = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

Utilizing Qdrant’s rapid metadata payload filtering, retrieval is scoped dynamically by semester week, curriculum topic, or document class, ensuring highly accurate responses and practically eliminating hallucinations. On-edge hardware, avoiding expensive semantic chunking strategies is computationally vital. Research by Anonymous (2024) confirms that simple static windowing with overlap on edge hardware is more computationally efficient and significantly easier to audit for data protection compliance than complex dynamic semantic boundaries.

A practical example of a Qdrant metadata payload filter looks as follows:

```
{
  "filter": {
    "must": [
      { "key": "course_id", "match": { "value": "information_systems_2026" } },
      { "key": "lecture_week", "match": { "value": 3 } },
      { "key": "document_class", "match": { "value": "syllabus" } }
    ]
  }
}
```

This metadata constraints the database query to the third lecture week of the specific 2026 course, preventing accidental retrieval of outdated or contextually mismatched course materials from other modules.

Parsing multi-column layout academic papers, embedded tables, and equations represents a crucial bottleneck for local RAG pipelines. While naive text chunkers fail to capture layout hierarchies, causing garbled context injections into the LLM, TRINITY leverages specialized layout-sensitive parsers. Docling (IBM Research 2024) identifies layout cells and converts nested tables into clean HTML or markdown tables. Mathematically dense sections are processed by Marker (Paruchuri 2023) to reconstruct LaTeX equations. This ensures that game-theoretic payoff matrices or formulas are represented exactly as intended.

Document chunks are processed using a static windowing approach (e.g., 512-character chunks with a 128-character overlap). Multilingual embeddings are computed using the paraphrase-multilingual-MiniLM-L12-v2 model, projecting texts into a 384-dimensional space. The search matches vector similarity via cosine distance. Qdrant's payload filters restrict search scopes to specific course IDs or teaching weeks, eliminating cross-context contamination and ensuring that only syllabus-grounded facts enter the model prompt, practically eradicating hallucinations.

2.4 Secure WebAssembly Python Sandbox

To enable mathematical calculations, data plotting, or interactive simulation generation directly from spoken commands, TRINITY provides a secure, sandboxed runtime environment. The desktop GUI, implemented via PySide6, instantiates a native QWebEngineView that loads **Pyodide**, a complete Python runtime compiled to WebAssembly (WASM).

When an agent (such as the `python_sandbox_agent`) generates Python code to plot mathematical functions, the code is not executed on the host's native operating system. Instead, it is injected as a string into the WebAssembly sandbox. It runs in a highly isolated environment with no access to local files or network sockets. The Pyodide runtime isolates system operations using Emscripten's virtual in-memory file system (MEMFS), ensuring that escaping the sandbox is structurally impossible. The resulting plots and HTML5 widgets are rendered inside the secure PySide6 interface, offering a clean, interactive, and safe computing experience.

3. Didactic Framework & Socratic Assistance

TRINITY’s pedagogical design is grounded in cognitive psychology and educational science. It departs from simple information retrieval, operationalizing the concepts of *pedagogical scaffolding* and *Complementary Learning Systems (CLS)*.

3.1 Cognitive Grounding: CLS Theory

TRINITY’s information processing structure mirrors human memory consolidation as described by CLS theory (McClelland et al. 1995; Kumaran et al. 2016). The human brain utilizes two complementary systems: 1. The **Hippocampus**, which rapidly encodes new episodic experiences in a highly specific, sequential manner. 2. The **Neocortex**, which slowly consolidates this raw episodic memory into structured, generalized conceptual schemas (often during offline replay).

TRINITY adapts this dual-system architecture: - **Episodic Memory System**: The active transcription stream in `core/transcriber.py` acts as the hippocampus, recording the sequential, raw timeline of the classroom session. - **Consolidated Knowledge System**: During classroom breaks or asynchronous intervals, a background thread (the Heartbeat Agent) summarizes raw transcripts, aligns them with pre-loaded syllabus structures, and updates the neocortical representation in the local Qdrant database.

This biologically inspired division of labor prevents the “catastrophic forgetting” phenomenon common in continuous streaming architectures and maximizes the value of domain-specific vector indexes in higher education (Li et al. 2025).

3.2 Human-Centered AI & Socratic Scaffolding

In contrast to central cloud systems that risk replacing or marginalizing the teacher, TRINITY adheres to the core guidelines of *Human-Centered Pedagogy* (ACUE 2024; Educause 2024). The instructor remains the exclusive pedagogical focal point in the classroom. TRINITY operates as an offline, silent assistant, offloading cognitive tasks to free up the instructor’s attention for direct student engagement.

The system supports this with three key didactic strategies: 1. **Socratic Prompts**: TRINITY monitors the lecture transcript in the background and generates open-ended questions or opposing viewpoints. These are fed to the professor’s earpiece to stimulate classroom discussion (Socratic Dialogue, cf. Goel & Polepeddi 2020). 2. **Cognitive Scaffolding**: During complex discussions, the system displays key definitions or historical context blocks. This offloads active recall tasks from the teacher’s working memory, allowing him to concentrate on classroom social dynamics. 3. **Truth-Anchored QA Generation**: Inspired by inverse QA techniques (ChartVerse Team 2025), TRINITY automatically generates review questions at the end of key lecture segments. These questions are strictly grounded in the spoken lecture transcript, ensuring absolute factual correctness, and can be pushed to students via Telegram or a project screen.

Concrete Interaction Scenario

To illustrate the socratic scaffolding process in a live lecture setting, consider the following scenario: * **Instructor says**: “...and if both actors act rationally, they will choose the Nash equilibrium, even though cooperative behavior would be collectively superior.” * **TRINITY transcribes**

this utterance in real time. * **The syllabus_agent identifies** the concept of “Nash Equilibrium” and matches it with the current week’s curriculum files. * **The definition_agent generates** a Socratic prompt and whispers it to the instructor’s earpiece: * *“Socratic prompt available: Ask students to apply the Nash Equilibrium concept to global climate agreements. What does the payoff matrix look like?”* * **The instructor incorporates this prompt naturally:** “Let’s illustrate this. Think about global climate agreements. How can we model the decisions of individual nations as a Nash equilibrium that conflicts with collective global survival?”

This sequence highlights how TRINITY enhances classroom discussion dynamically, keeping the professor in control of the learning experience without introducing distracting digital barriers.

In pedagogical theory, Socratic scaffolding is closely aligned with Vygotsky’s “Zone of Proximal Development” (ZPD). The AI acts as a temporary cognitive support, offloading active retrieval tasks from the instructor so they can focus on high-order moderation and student interactions. The instructor does not need to allocate working memory to recall exact historical dates or niche definitions—TRINITY whispers them directly. This structural cognitive offloading (Cognitive Load Theory) dramatically reduces the instructor’s mental fatigue during extended lecturing sessions.

Furthermore, TRINITY’s truth-anchored QA generator (ChartVerse Team 2025) ensures that review questions generated at the end of a lecture are strictly tied to what was actually discussed in the room. This aligns perfectly with John Biggs’ “Constructive Alignment” framework, linking learning objectives, live delivery, and assessment in a unified, verified loop, completely bypassing the hallucination risks common in general-purpose cloud generators.

4. User-Centered Workflows & AirPods Interface

TRINITY’s user experience (UX) is optimized to fit seamlessly into the active workflow of university faculty. The system defines three specialized modes managed through a highly intuitive, distraction-free interface.

4.1 The Three Operating Modes (Lecture, Office, Chat)

- **Lecture Mode:** Designed for hands-free operation. The instructor moves freely around the hall, talking naturally, and invokes digital actions or visual slides exclusively via the fuzzy wake-word “Trinity.”
- **Office Mode:** Streamlines administrative tasks. Through a clean drag-and-drop interface in the desktop PySide6 app, professors upload student milestone drafts. TRINITY performs layout-sensitive parsing and outputs structured grading recommendation drafts (Anonymous 2026).
- **Chat Mode:** A classic, premium chat UI featuring responsive glassmorphic widgets, used for deep literature reviews and system configuration.

4.2 AirPods Audio Routing (Silent Prompting)

A vital UX design is the physical audio split coordinated through the macOS CoreAudio subsystem: - By default, the system speaks (TTS using macOS native say engine) quietly and exclusively into the professor’s AirPods (Silent Prompting). The professor receives cues and facts without interrupting the classroom’s audio. - When the agentic router detects an explicit public command (e.g., “*Trinity, display this simulation and explain it to everyone*”), it dynamically switches

CoreAudio routing to the primary system output (classroom speakers), making the explanation audible to the entire room.

4.3 Heartbeat Background Analysis & Telegram Bridge

During slides-driven lectures, interrupting the presentation with UI popups would be highly distracting. TRINITY addresses this through an asynchronous communication loop: The background_heartbeat_loop in core/transcriber.py runs every 2 minutes. It analyzes the gathered transcript for logical contradictions, pacing issues, or missing curriculum points. When a warning is generated, TRINITY sends it silently as a Telegram direct message to the professor's phone or smartwatch, allowing him to adapt his lecture on the fly without the audience noticing.

These interactions and their relative architectural dimensions are illustrated in the following structural research map:



Figure 2: Figure 2: Project Trinity Structural Research Map and Dimension Weights

5. Privacy Design & GDPR Compliance

Deploying AI in European educational institutions requires strict adherence to the General Data Protection Regulation (GDPR). Since TRINITY captures real-time audio and processes sensitive academic files, it is architected strictly under the principles of **Privacy by Design** (Art. 25 GDPR).

The system employs a defensive legal stance, relying on physical, structural, and local processing boundaries.

5.1 Physical Microphone Attenuation Argument

A prominent legal obstacle for real-time speech systems in public spaces is securing explicit recording consent from everyone in the room. If a broad room-mic is used, students' voices are captured, raising severe compliance concerns.

TRINITY solves this via a physical hardware boundary: Audio is captured solely through a single wireless AirPods worn by the instructor. AirPods feature a highly directional near-field mic design, engineered to capture audio only within a close proximity boundary of maximum 20 cm from the wearer's mouth. Ambient sounds originating from further away – such as student questions from the seating rows – are physically attenuated by more than 40 dB.

Mathematically, the sound pressure level L_p as a function of the distance r from the microphone, considering the inverse square law and the microphone's polar directivity factor, is modeled as:

$$L_p(r) = L_{p0} - 20 \log_{10} \left(\frac{r}{r_0} \right) - D(\theta)$$

where $D(\theta)$ is the attenuation in decibels at off-axis angle θ . For ambient student speakers situated at distances $r \geq 3$ meters and off-axis angles $\theta \geq 45^\circ$, the accumulated sound attenuation physically suppresses the signal, burying it in the room's ambient noise floor.

This physical sound damping ensures that capturing or identifying student voices is structurally impossible. The audio stream contains only the instructor's spoken words.

5.2 Local Data Residency & Data Minimization

Adhering to data minimization principles (Art. 5(1)(c) GDPR), all sensitive user and student data is processed exclusively on the local machine: - **On-Device Inference:** Speech transcription (faster-whisper), embedding computation (sentence-transformers), and model inference (LiteRT-LM, MLX) run fully offline on Apple Silicon NPUs and GPUs. No raw audio or text is transmitted to commercial cloud servers. - **Secure Local Storage:** The Qdrant vector database and session logs are stored locally and encrypted on the device's storage.

This decentralized approach practically eliminates the risk of external data breaches, fully aligning with UNESCO recommendations on digital sovereignty in education (UNESCO 2024; DEC 2024).

[!IMPORTANT] While TRINITY's architecture is designed to support a robust data protection argument, it does not guarantee automatic GDPR compliance. Educational institutions must perform their own formal Data Protection Impact Assessment (DPIA) with their designated privacy officer prior to any commercial or widespread institutional deployment.

The physical microphone attenuation argument is a cornerstone of TRINITY's compliance strategy under European data laws. In many EU member states, unauthorized recording of spoken words is a criminal offence. Deploying ambient classroom microphones would legally require

written consent from every student present—an administrative impossibility. TRINITY circumvents this challenge through physical hardware boundaries.

The AirPods’ dual beamforming microphone array leverages phase-delay filtering and arrival-time differentials between its physical ports to create an extremely narrow directional polar pattern. Acoustic sources originating outside a 20 cm distance threshold from the wearer’s mouth are attenuated by more than 40 dB via hardware-level destructive interference. The directivity factor $D(\theta)$ rises steeply for off-axis angles $\theta > 30^\circ$. Consequently, student questions, ambient room noise, or rustling papers from the lecture rows are physically suppressed, rendering them unreadable by the local speech-to-text pipeline.

This physical boundary is complemented by strict local data residency. No raw audio, transcript files, or RAG vectors ever leave the host device. All model execution (Whisper, Gemma 4, Qwen 0.8B) occurs fully offline on the local processor, satisfying the core GDPR principles of data minimization and security.

6. Technical Evaluation & Benchmarks

To validate TRINITY’s real-world feasibility, we conducted several technical benchmarks measuring voice processing latency and mobile RAM consumption.

6.1 Voice Processing Latency

A natural lecturing flow demands immediate responses. The total latency T_{total} consists of capture buffering (t_{cap}), speech-to-text (t_{stt}), agentic routing (t_{route}), model generation (t_{llm}), and text-to-speech (t_{tts}):

$$T_{\text{total}} = t_{\text{cap}} + t_{\text{stt}} + t_{\text{route}} + t_{\text{llm}} + t_{\text{tts}}$$

The following table summarizes average latencies (over 100 test runs in Lecture Mode) recorded on various Apple Silicon platforms:

Test Platform / Device	STT Latency (t_{stt})	LLM Latency (t_{llm})	End-to-End Latency (T_{total})
MacBook Pro M3 Max (36 GB)	120 ms	180 ms	455 ms
MacBook Air M1 (16 GB)	210 ms	340 ms	720 ms
iPhone 15 Pro (8 GB, LiteRT)	180 ms	280 ms	610 ms
iPad Pro M2 (8 GB)	160 ms	240 ms	550 ms

These latency measurements prove that the local processing loop operates well below the conversational interruption threshold (approx. 1.5 seconds), guaranteeing seamless live support.

6.2 Mobile Memory Management (iOS Jetsam Limits)

Porting TRINITY to mobile devices (the *Souffleur* iOS app) presented significant RAM challenges. The iOS operating system incorporates a strict memory watchdog (Jetsam) that terminates applications exceeding a critical memory footprint. On an iPhone 15 Pro (8 GB RAM), the Jetsam threshold for background tasks is approximately 1.4 GB.

Initial tests running a 2-billion parameter model (Gemma 4 E2B .litertlm via native C++ XCFrameworks) under sustained loads caused memory peaks over 1.6 GB, triggering immediate Jetsam crashes.

TRINITY overcomes this by employing **hybrid mobile routing**: 1. **LiteRT-LM C++ Bridge**: For precise reasoning, Gemma 4 is executed via an optimized C++ runtime that strictly deallocates temporary tensors after each inference pass to minimize memory leaks. 2. **MLX Swift Fallback**: If the system RAM approaches 1.2 GB, the mobile router dynamically pivots to a lighter model – **Qwen 0.8B** (Alibaba Qwen Team 2025) quantized to 4-bit, executed via Apple’s native MLX Swift framework.

Qwen 0.8B maintains a memory footprint below 750 MB, ensuring stable, crash-free, and rapid (<50 ms time-to-first-token) performance even during extended lecture sessions.

The memory monitoring and switching logic inside the *Souffleur* iOS app is governed by the following Swift router structure:

```
// Swift-based hybrid model router for avoiding iOS Jetsam crashes
class MobileModelRouter {
    let memoryLimitMB: Double = 1200.0
    var activeEngine: InferenceEngine = .gemma4_LiteRT

    func routeQuery(_ query: String) -> String {
        let currentRAM = SystemMemory.getCurrentAppFootprintMB()
        if currentRAM > memoryLimitMB && activeEngine == .gemma4_LiteRT {
            print("Critical memory pressure detected (\(currentRAM) MB).
Switching to Qwen 0.8B fallback...")
            activeEngine = .qwen08B_MLX
        } else if currentRAM < 900.0 && activeEngine == .qwen08B_MLX {
            print("Memory footprint stabilized (\(currentRAM) MB). Switching back
to Gemma 4 LiteRT...")
            activeEngine = .gemma4_LiteRT
        }
        return activeEngine.execute(query)
    }
}
```

This reactive routing layer safeguards the application from unexpected shutdowns, ensuring the “silent concierge” remains available throughout the entire academic session.

Our technical latency benchmarks show that local inference loops on Apple Silicon deliver exceptional real-time responsiveness. On a MacBook Pro M3 Max, the entire loop from voice input to synthetic audio playback requires only 455 ms. This is far below the human conversational gap

threshold (approx. 1.5 seconds). Even on budget-oriented hardware like a MacBook Air M1, end-to-end latency remains below 720 ms, which is highly practical for live auditorium support.

A major challenge when deploying to mobile devices (the Souffleur iOS app) was navigating the iOS Jetsam memory manager. iOS enforces aggressive memory limits on applications to protect system responsiveness. On a typical 8 GB iPhone, background or foreground apps exceeding approximately 1.4 GB are immediately terminated by the OS. Running a full 2-billion parameter model like Gemma 4 E2B under sustained loads would peak at 1.6 GB, causing recurrent app crashes. TRINITY bypasses this using an active hybrid model router:

Primarily, the system runs Gemma 4 E2B (Google 2026) via a native, optimized LiteRT-LM C++ wrapper, which minimizes memory overhead through direct tensor deallocations. If the system memory footprint approaches 1.2 GB, the Swift router seamlessly redirects incoming requests to a highly compact fallback engine: the 4-bit quantized **Qwen 0.8B** model (Alibaba Qwen Team 2025) running on Apple’s native MLX Swift framework. The Qwen engine operates stably below 750 MB RAM, completely preventing Jetsam shutdowns and maintaining continuous, zero-interruption auditory support.

7. Discussion & Future Work

TRINITY demonstrates that decentralized, local edge AI is a highly performant and pedagogically superior alternative to centralized cloud APIs. Nonetheless, certain physical and algorithmic limitations remain.

7.1 Bounds of Local Architectures

While local Apple Silicon execution guarantees low latencies, processing extremely large context windows is bound by physical memory bandwidth. Over multi-hour lecture series, accumulated transcripts grow rapidly. Since unified memory is shared between computing cores, excessive context size degrades generation speeds. TRINITY mitigates this using recursive semantic summarization (Kumaran et al. 2016; Anonymous 2024), which, although highly effective, can occasionally omit minor historical nuances.

Additionally, highly intensive operations (like local ComfyUI image rendering or complex 3D math simulations) are currently offloaded to the host Mac, limiting full standalone functionality on iOS devices when completely offline.

7.2 Future Research Paths

Future work will explore the following directions: - **Multimodal Visual Inputs:** Interfacing with smart glasses (e.g., Apple Vision Pro) would allow didactical cues and parsed RAG data to be projected directly into the teacher’s visual field, reducing reliance on audio-only feedback. - **On-Device Speculative Decoding:** Integrating speculative decoding (Leviathan et al. 2023) – utilizing Qwen 0.8B as a draft model to speed up Gemma 4 verification – could double local generation speeds on mobile devices. - **Federated RAG Synchronization:** Implementing decentralized P2P synchronization of local Qdrant indexes would allow cross-faculty syllabus grounding without compromising academic data residency. - **Hardware-Level Kernel Optimization:** Recent breakthroughs like ThunderKittens (Stanford Hazy Research 2024) prove that

hardware-tile abstractions at register levels maximize memory throughput on GPUs. Integrating these techniques directly into the Metal/MLX execution kernels could enable running 8B or 14B models on consumer academic hardware in real time.

The constraints of offline edge architectures are closely bound to physical hardware limits. Although unified memory systems on modern M-series chips provide outstanding bandwidth between computing cores, total memory capacity bounds the maximum practical context window. In extended lectures, accumulated transcripts grow rapidly. Processing extremely long contexts on edge hardware inflates inference latency because the Key-Value (KV) cache computational cost scales quadratically with input length.

TRINITY addresses this using a two-tiered compression strategy. The raw transcript is periodically summarized and structured into markdown files by the `transcription_consolidator` skill. For real-time RAG, a rolling sliding window is utilized, keeping only the most recent audio chunks in active RAM, while historical contexts are written to the local Qdrant database.

Future work will target speculative decoding on mobile devices. By using a lightweight draft model (like Qwen 0.8B) to generate candidate tokens that are subsequently validated in a single parallel step by the target model (Gemma 4), local generation throughput on iOS could be doubled without compromising quality (Leviathan et al. 2023). Furthermore, integrating low-level register-tile abstractions like ThunderKittens (Stanford Hazy Research 2024) into Apple’s MLX and Metal execution kernels could dramatically boost GPU throughput, enabling fluid execution of larger 8B or 14B models on standard consumer hardware.

8. Conclusion

TRINITY represents a significant step forward in integrating artificial intelligence into higher education. The system moves away from centralized, monolithic chatbots, establishing instead the paradigm of a *local Academic Personal Concierge*.

By leveraging hardware-accelerated local execution (Apple Silicon, Metal, MLX Swift, and LiteRT-LM), TRINITY proves that speech processing, agentic routing, and vector retrieval can be performed entirely offline with exceptional performance. Pedagogically, the system strengthens the physical presence and autonomy of the teacher. Legally, the physical AirPods near-field directional microphone argument combined with local data residency offers a pragmatic path forward under GDPR constraints.

TRINITY provides a robust blueprint for a new class of secure, decentralized, and human-centered AI systems that build lasting trust in academic technology.

Appendix: Technical Specifications and Integration Details

A.1 Initialization of the Silero VAD Ring Buffer

The software-level interface to CoreAudio and the Silero VAD module in `core/transcriber.py` is implemented using the following Python audio buffer configuration:

```

# Initialization of the asynchronous audio buffer for Silero VAD
import numpy as np
import collections

class AudioBuffer:
    def __init__(self, sample_rate=16000, window_size_samples=512):
        self.sample_rate = sample_rate
        self.window_size = window_size_samples
        self.buffer = collections.deque(maxlen=100) # stores up to 3.2 seconds of
audio
        self.triggered = False

    def append(self, frame):
        self.buffer.append(frame)

    def get_audio_segment(self):
        return np.concatenate(list(self.buffer), axis=0)

```

This ring buffer configuration prevents input queue overflow and ensures a constant, jitter-free feed of audio frames for speech activity classification.

A.2 Structured MCP Tool Definition Schema

The following JSON schema represents how specialized skills such as the `slides_agent` define their input expectations to the local MCP orchestrator:

```

{
  "name": "slides_agent",
  "description": "Orchestrates presentation slide transitions based on spoken
commands",
  "inputSchema": {
    "type": "object",
    "properties": {
      "command": {
        "type": "string",
        "description": "The target transition: 'next', 'previous', or
'go_to_slide'"
      },
      "target_slide_title": {
        "type": "string",
        "description": "Optional title string matching the slide header for
direct navigation"
      }
    },
    "required": ["command"]
  }
}

```

This structured format ensures that the local agentic brain in `core/brain.py` executes exact validation and parameter safety checks before dispatching operating system tasks.

A.3 Methodological Generation Process and AI Agent Orchestration

This document was generated entirely using an agentic AI system as part of a structured, multi-phase research and writing process.

Language Model Used

- **Model:** Gemini 3.5 Flash (Medium Thinking)

Specialized AI Agents (Roles & Tasks)

1. **academicwriting_agent:** Responsible for the overall structure of the paper in academic style, writing in the “Engel style,” maintaining bilingual symmetry, conducting peer review iterations (“Richter rounds”), and enforcing the minimum word count of over 7,500 words.
2. **deep_research_citation:** Managed the literature database (`literatur.json`), compiled the source-based evidence matrix, generated the research sources registry, and performed the in-text citation audit to prevent phantom references and hallucinated DOIs.
3. **imagegeneration:** Constructed the detailed JSON prompt for the conceptual architecture diagram of TRINITY and executed the premium image generation via the Kie.ai API.
4. **defuddle (optional):** Cleaned raw data extracted from web pages to discard boilerplates and navigation links for RAG searches.

Execution Phases

- **Phase 1: Topic and Boundaries:** Aligned the research scope, formulated the core thesis, and established the contribution profile (35% Tech/Architecture, 30% Didactics, 25% UX/Workflow, 10% Legal/GDPR).
- **Phase 2: Internal Deep Research:** Analyzed the Trinity codebase and configuration files across the desktop and mobile repositories, resolving iOS memory (Jetsam) thresholds.
- **Phase 3: External Deep Research:** Identified, analyzed, and verified 24 academic, technical, and policy sources.
- **Phase 4: Research Map & Paper Outline:** Structured the 9 core research kernels, mapped citations, and created the vector-based research map (`media/research_map_bubbles.svg`).
- **Phase 4.5: Kie.ai Diagram Generation:** Automatically generated the conceptual system architecture diagram (`media/trinity_architecture.jpg`).
- **Phase 5: Paper writing:** Bilingual text generation, followed by resolving LaTeX backslash escaping issues using Python raw strings (r-strings).
- **Phase 6: Final Acceptance Audit:** Verified word counts, citation coverage, and compiled PDFs to validate overall compliance.

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Synopsis: This technical report details the highly efficient Qwen model family, highlighting the reasoning capabilities of ultra-compact models (e.g., Qwen 0.8B) in structured JSON extraction and logical inference tasks under constrained RAM footprints, validating its selection as TRINITY’s mobile fallback engine.
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Synopsis: The authors evaluate the computational overhead of semantic chunking compared to static overlapping windows on edge devices. The paper provides the scientific basis for TRINITY’s choice of static overlapping window chunking, which guarantees high processing throughput and predictable compliance auditing on local hardware.
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Synopsis: Documents the pure C/C++ port of Whisper, highly optimized for Apple Silicon (Metal, Accelerate) for low-latency offline transcription, which forms the foundation of TRINITY’s audio transcription layer.
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